Sampling Streets: Design-Based Inference for Street-Level Field Measurement

Varun K.R. Gaurav Sood*

November 2026

Abstract

Researchers measure streets with cameras and sensors: the condition of roads, who is visible in public, how loud and how polluted the street is. The data usually come from wherever access was easy, so the numbers describe the places measured and not the city. We describe what it takes to make such a measurement a statement about a city: a list of road segments to draw from, a random draw, and standard errors that follow from the draw. We then show that drawing points at random is not enough. When the observed unit is the drawn unit, as when a Street View image is requested at a sampled segment, the design carries over, and what needs care is coverage and measurement. When a field team rides between sampled points with a camera or a meter, the observations describe the route, not the sample: routes run on bigger roads than the points sit on, and a camera cut at fixed time intervals photographs slow stretches more often. In Delhi, 85% of sampled points but 27% of video frames are on residential roads. Reweighting road classes to the drawn sample, with frames weighted by speed, moves the estimated share of women among pedestrians in Delhi from 20.4% to 23.5%, a shift of 3.1 points (95% interval 0.5 to 6.8). Reweighting road classes moves the share of Delhi noise readings peaking at or above 100 dB from 14.4% to 8.1%. In a simulation where road class is the only way routes differ from the city, intervals from the unadjusted route sample cover the truth 32% of the time and the reweighted ones 95%. We show that a route sample is still a probability sample: a stretch of road is photographed at a rate proportional to how often the routing rule sends the rider along

*Data and code to reproduce every number in this paper are at https://github.com/geosensing/streetscan. The sampler, the route allocator, and the estimation package are at https://github.com/geosensing.

it, divided by the rider's speed there. Weighting by the inverse of that rate is unbiased, and reweighting by road class is its approximation when the rate is taken as constant within a class. Within road classes, frames beside sampled points and frames on the way between them show similar shares of women. We apply the framework to road condition in four cities, women's presence on the street in four Indian cities, and noise and air quality in Delhi.

1 Introduction

Measurements taken on the street are an increasingly common source of data in the social sciences. Researchers code neighborhood disorder, road and sidewalk condition, and the composition of public space from street-level imagery [Rundle et al., 2011, Kelly et al., 2013, Naik et al., 2014, Gebru et al., 2017], and they measure noise, particulate matter, and temperature with sensors. Some of this work observes a designed sample of street segments; Sampson and Raudenbush [1999] filmed more than 23,000 Chicago block faces from a vehicle. Much of it draws on data that are convenient: a platform's image corpus, a census of named blocks, monitors at fixed stations, or routes a team could reach. Convenient data deliver volume. They do not say what population the numbers describe, and standard errors computed from them treat observations as independent draws from an unspecified population.

Survey sampling has a standard answer. List the units of the population, draw some at random with known probabilities, and weight each observation by the inverse of its probability of being drawn [Horvitz and Thompson, 1952, Cochran, 1977, Lohr, 2021]. For streets, the list can be built from the road network: OpenStreetMap roads [OpenStreetMap contributors, 2026] cut into short segments. We built tools to do this: a sampler that cuts roads into segments and draws from them, an allocator that groups sampled points into routes a field team can ride in a day, and an estimation package.

This paper makes two points. The first is that the standard machinery applies directly, and easily, when the unit observed is the unit drawn. If an image is requested at the start of each sampled segment, each image stands for the segments the draw says it stands for. The remaining problems are coverage, since some sampled segments have no image, and measurement, since one image stands for a segment.

The second point is a correction to our own practice. When a field team rides between sampled points with a camera that is always recording, or stops to read a meter where it is convenient to stop, the observations are not at the sampled points. The sample fixes a route, and the data describe the route. A route differs from the road network in two ways that the

data can show and partly repair. Routes between points run on bigger roads than the points sit on. And a camera that saves a frame every few seconds photographs a stretch of road in proportion to the time spent on it, so slow stretches, which are also the crowded ones, are over-represented. Neither fact is new. That averages over time and over distance differ by a speed weighting is a standard result in traffic theory [Wardrop, 1952], and reweighting a sample to known class shares is post-stratification [Holt and Smith, 1979]. Our contribution is to show that a route sample is nonetheless a probability sample with a rate of inclusion that can be written down, to state what each estimator converges to, to check this in simulations with a known truth, and to show how far the two facts separate route data from a drawn sample in practice.

We use five applications. Road condition in Bangkok, Jakarta, Lagos, and Wayne County, Michigan, comes from Street View images at sampled segments. Women's share of pedestrians comes from Street View at sampled locations in Mumbai, Navi Mumbai, and Delhi, and from video recorded on a motorbike in those cities and Bangalore, an outcome motivated by the concern that women are underrepresented in everyday public life [Sen, 1990, Phadke et al., 2011]. Noise and air quality in Delhi come from hand-held meters read along the same routes; road-traffic noise affects sleep, cardiovascular health, and cognition [World Health Organization. Regional Office for Europe, 2018, World Health Organization, 2022].

2 Estimands

Let $i \in \mathcal{U} = \{1, \ldots, N\}$ index the units of a finite population, here segments of the road network. For unit i, Y_i is a measurement and, where relevant, Z_i a denominator. For counts of people by gender, Y_i is the number of women visible and Z_i the number of people classified as women or men. For a sound-meter reading, Y_i is a level in decibels and there is no denominator.

The *population ratio* is

$$R = \frac{\sum_{i \in \mathcal{U}} Y_i}{\sum_{i \in \mathcal{U}} Z_i}. \tag{1}$$

For images, R is the share of visible people who are women. A market with two hundred people counts a hundred times as much as a side street with two.

The *population mean* is $\mu = N^{-1} \sum_{i \in \mathcal{U}} Y_i$: for noise, the sound level at a segment drawn at random. The *threshold share* is

$$P_\tau = \frac{1}{N} \sum_{i \in \mathcal{U}} \mathbb{1}\{Y_i \geq \tau\}, \tag{2}$$

the share of segments at or above a level τ. For noise we use $\tau = 55$ dB, the daytime ambient standard for residential areas under India's Noise Pollution Rules [Ministry of Environment and Forests, Government of India, 2000], for the quietest moment in a reading, and $\tau = 100$ dB for the loudest.

A fourth quantity, the *location mean* of a per-unit ratio,

$$L = \frac{1}{N^+} \sum_{i \in \mathcal{U}: Z_i > 0} \frac{Y_i}{Z_i}, \tag{3}$$

counts a side street the same as a market. R below L would mean that crowded places are more male.

All four are defined over segments. Whether data estimate them depends on how the data were collected, which is the subject of the next section. A camera that saves a frame every minute estimates what a rider sees per minute. That is a well-defined quantity, but it is not R.

3 Two Designs

3.1 The frame

The frame is a list of road segments. Roads of chosen classes are taken from OpenStreetMap and cut into pieces of about 500 meters; each piece is a row. The frame says what the estimate is about. A frame of primary, secondary, tertiary, and trunk roads supports claims about major roads. A frame that adds residential streets supports claims about the network people live on, most of which is residential: in the samples drawn for the video fieldwork, residential streets are 60 to 85% of points.

The simplest draw takes n segments from N with equal probability, so each has inclusion probability $\pi_i = n/N$. A stratified draw takes n_h of the N_h segments in each road class h, so $\pi_i = n_h/N_h$.

3.2 Point samples

In a point sample the unit observed is the unit drawn. The clearest case is platform imagery. Google Street View's metadata service says whether a panorama exists near a coordinate and when it was taken, and its image service returns the picture [Google, 2026b,c]. Query it at the start of each sampled segment, and each image inherits its segment's π_i.

Two things can go wrong. The first is coverage. Some sampled segments have no panorama, and not every available image gets coded. If the missing segments differ from

the rest, estimates describe covered streets only. Availability is known to vary with the wealth of an area [Fry et al., 2020]. The share of sampled segments with an image is itself an estimate of the platform's coverage of the frame, and it should be reported. The second is measurement. The service returns a panorama near the coordinate, not at it; one view stands for a 500-meter segment; and coders err. None of this is a sampling problem, but all of it sits between the image and the segment's true value.

3.3 Route samples

Sending a person to each sampled point is expensive, so sampled points are grouped into itineraries and a rider travels each itinerary. In our fieldwork the rider wore a camera that recorded continuously, and frames were later saved from the video at fixed intervals: every 10 seconds in Mumbai and Navi Mumbai and every 120 seconds in Bangalore and Delhi. In Mumbai and Navi Mumbai a random subset of the saved frames was annotated: 25% of 10,867 in Mumbai and 36% of 4,738 in Navi Mumbai. A random subset of frames spaced evenly in time is still spread evenly in time. The script that drew the subset is not in the repository, so we checked the result. Annotation did not favor the start or the end of a ride: the annotated share is 22% in the first half of a video's frames and 21% in the second half in Mumbai, and 36% and 37% in Navi Mumbai. In Navi Mumbai the annotated share does not vary across videos more than chance allows. In Mumbai it does: 6 of 125 videos were annotated in full and 11 not at all. No frame is tied to a sampled point. The median frame is 68 meters from the nearest sampled point in Mumbai, 368 in Delhi, and 450 in Bangalore. Sound-meter readings were taken at stops along the route, a median 112 meters from the nearest sampled point, and 45% of them within 100 meters of one.

The sampled points therefore fix the route, and the observations are a sample of the route. Two things separate them from the frame.

Time. A frame every t seconds lands on a given stretch of road with probability proportional to the time the rider spends there, which is the stretch's length divided by the rider's speed. A stretch ridden at 5 km/h is photographed five times as often as one ridden at 25 km/h. Speed is lowest where streets are narrow or crowded, so the frames over-represent those places relative to distance. In Delhi the median speed at a frame is 8 km/h on residential roads and 26 km/h on secondary and larger roads, and in 14% of frames the rider is stationary. Weighting each frame by the rider's speed at that moment, v_i, turns a sample that is uniform in time into one that is uniform in distance:

$$\widehat{R}_{\text{route}} = \frac{\sum_{i \in \mathcal{S}} v_i Y_i}{\sum_{i \in \mathcal{S}} v_i Z_i}. \tag{4}$$

Speed comes from the camera's GPS track. A stationary frame would get no weight, so we floor speed at 0.5 meters per second.

Roads. The quickest way between two points on residential streets usually runs along a larger road. Figure 1 compares the road classes of the sampled points with those of the frames. In Delhi, 85% of sampled points and 27% of frames are on residential roads; 4% of points and 45% of frames are on secondary or larger roads. Weighting by speed makes this worse, not better, because it shifts weight toward fast roads. The repair is post-stratification [Holt and Smith, 1979]: rescale the weights so that each road class has the share it has in the frame. We use four classes: residential, tertiary, secondary and larger, and other. The drawn sample estimates the frame's shares, because every draw here gave segments equal probability (in Mumbai, by allocating the sample to road classes in proportion to their size). Frames on roads of a class absent from the drawn sample are outside the frame and are dropped.

This removes the imbalance across road classes. It cannot remove differences within a class between the streets a route uses and the streets it does not. Section 4 makes this exact: a route sample has a rate of inclusion that the design determines, and post-stratification replaces that rate with its average in each road class.

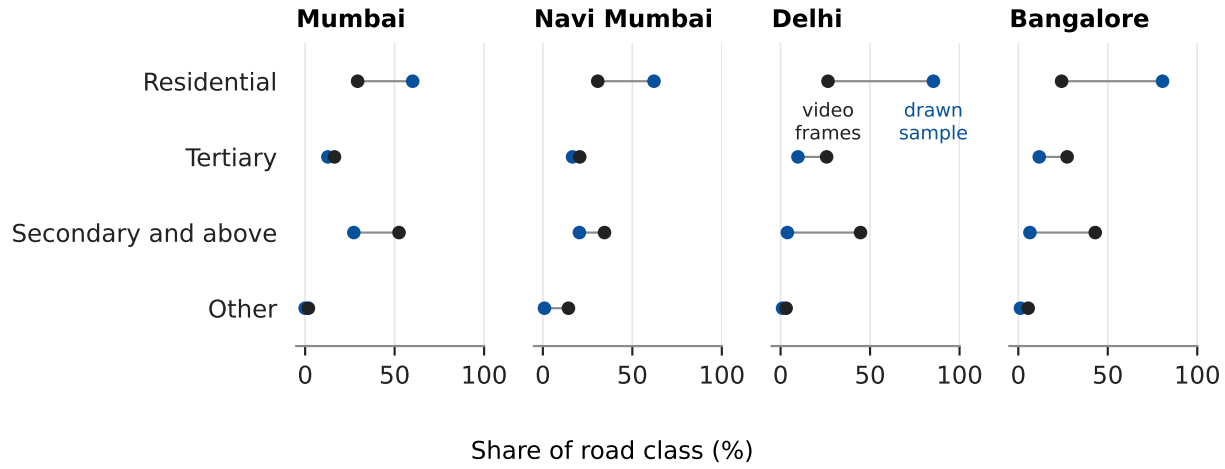


Figure 1: Road classes of the sampled points and of the video frames recorded while riding between them. Each panel is a city. Blue points give the share of the sampled points in each OpenStreetMap road class; black points give the share of annotated video frames, classed by the nearest mapped road. "Other" includes service roads, footways, unclassified roads, and frames with no mapped road nearby.

The rider also did not complete every itinerary. Video names identify 52 of Delhi's 85 itineraries and 50 of Bangalore's 111, and we do not know how those were chosen.

Which of the three quantities is wanted depends on the question. Per minute of riding is the rider's experience. Per meter of route describes the roads ridden. Reweighted to the

frame is the closest a route sample comes to the road network.

3.4 Allocation

Grouping sampled points into itineraries does not change which points were drawn. It does make observations depend on one another: frames from one day share weather, a time of day, a part of the city, and a rider. Standard errors must treat the day or the itinerary, not the frame, as the independent unit.

Our allocator can also generate random walks on the road graph in place of routes between sampled points. A simple random walk on an undirected graph crosses every edge equally often in the long run, so measurements taken at a fixed distance along it would weight roads by length without any reweighting. That standard property of random walks is a limit: it holds for long walks on a connected network and fails on one-way streets. None of the applications here used walks, and we do not evaluate them.

4 A Route Sample Is a Probability Sample

A route sample looks like a sample of convenience, since the rider goes where the road leads. It is not. The rider goes where a rule sends them, and the rule's input is a random draw. That is enough to give every stretch of road a rate of being photographed that follows from the design.

4.1 Setup

Let the network be a set of roads with total length Λ, and write $\int g(x)\,dx$ for an integral over road length. Let $y(x)$ and $z(x)$ be what the camera would record at location x. A design draws a set of points S. A routing rule turns S into a ride. Let $c_S(x)$ be the number of times the ride passes x, and

$$m(x) = \mathbb{E}_S\left[c_S(x)\right] \tag{5}$$

its average over draws, the *traversal intensity* of x. The rider's speed at time t is $v(t)$. The camera saves a frame at times $U, U+\Delta, U+2\Delta, \ldots$, where the start U is uniform on $[0, \Delta]$. Frame i is at location x_i and speed v_i.

We assume four things. (A1) The ride is a function of the draw alone, or departs from the rule in ways unrelated to y and z. (A2) The timer's start is independent of the draw and the ride. (A3) y and z are properties of the place over the fieldwork window, not of the

moment the rider passes. (A4) $m(x) > 0$ wherever the estimand is defined: the rule can send the rider along every street.

4.2 Results

Proposition 1 (Rate of inclusion)**.** *Under A1 and A2, for any bounded g,*

$$\mathbb{E}\Big[\sum_i v_i\, g(x_i)\Big] = \frac{1}{\Delta}\int m(x)\, g(x)\, dx. \tag{6}$$

If speed depends only on location, $\mathbb{E}\Big[\sum_i g(x_i)\Big] = \Delta^{-1}\int (m(x)/v(x))\, g(x)\, dx$: frames fall on a stretch at a rate proportional to m/v.

Proof. Fix the draw, so the ride is a path $x(t)$ for $t \in [0, T]$. For any integrable h, a timer with a uniform start satisfies $\mathbb{E}_U \sum_k h(U + k\Delta) = \Delta^{-1}\int_0^T h(t)\, dt$. Take $h(t) = v(t)\, g(x(t))$. Distance along the ride is $ds = v(t)\, dt$, so $\int_0^T v(t)\, g(x(t))\, dt = \int g\, ds$, the integral of g along the ride, which counts each location once per pass: $\int c_S(x)\, g(x)\, dx$. Averaging over draws replaces c_S by m. For the second statement take $h(t) = g(x(t))$ and $dt = ds/v$. □

Time spent stationary contributes nothing to the first sum, because $v = 0$ there. The floor on speed in Section 3 is therefore a convenience, not a requirement.

Corollary 1 (Design weights)**.** *Under A1, A2, and A4, $\Delta \sum_i v_i\, g(x_i)/m(x_i)$ is unbiased for $\int g(x)\, dx$.*

This is the Horvitz–Thompson estimator with frames weighted by speed over traversal intensity. The intensity m needs no model. It is a property of the draw and the routing rule, and it can be computed by drawing again and routing again many times [Fattorini, 2006].

Proposition 2 (What each estimator converges to)**.** *Suppose rides are independent draws from the design and A1 to A3 hold. As the number of rides grows, the ratio of weighted sums $\sum_i w_i y_i / \sum_i w_i z_i$ converges as follows.*

1. *With $w_i = 1$ and speed a function of location, to $\int (m/v)\, y \,/\, \int (m/v)\, z$.*
2. *With $w_i = v_i$, to $R_m = \int m\, y \,/\, \int m\, z$.*
3. *With $w_i = v_i/m(x_i)$, under A4, to $R = \int y \,/\, \int z$.*
4. *With $w_i = v_i$ post-stratified to classes c with length shares λ_c, to $\sum_c \lambda_c \bar{y}_c \,/\, \sum_c \lambda_c \bar{z}_c$, where $\bar{y}_c = \int_c m\, y / \int_c m$ is the mean of y in class c weighted by m. This equals R when, within every class, m is uncorrelated with y and with z, and in particular when m is constant within classes.*

Proof. By Proposition 1, each ride's numerator and denominator have the expectations whose ratio is the stated limit. Rides are independent and identically distributed, so the averages of numerators and of denominators converge to those expectations, and so does their ratio. For the fourth case, the speed-weighted sum of y in class c divided by the speed-weighted count converges to $\bar{y}_c$, and post-stratification combines classes with weights λ_c. Within a class, $\bar{y}_c$ minus the plain mean of y is the covariance of m and y in the class divided by the mean of m, which is zero under the stated condition. □

For a mean ($z = 1$), the second case gives the bias of speed weighting alone: the covariance of m and y over the network, divided by the mean of m. So a route sample needs only speed weights exactly when the streets the rule favors are no different, in the outcome, from the streets it avoids. The first case shows something else. The timer over-samples slow streets and the routing rule over-samples fast ones, so the two distortions pull in opposite directions, and where m/v is roughly constant they cancel and the unweighted estimate is close to right by accident.

Three limits apply to our fieldwork. First, we did not compute m. The rides followed a navigation application whose choices we cannot replay, and not every itinerary was ridden. We use post-stratification, which is the fourth case: it takes m as constant within a road class. Second, consistency here is in the number of rides, and we have 9 to 13 days per city. Third, the result covers a timer. Sound-meter readings were taken where the rider chose to stop, which is not a rule we can write down.

A random subset of frames, as in Mumbai and Navi Mumbai, changes nothing: keeping each frame with a fixed probability multiplies every rate by that probability, which cancels in a ratio.

4.3 A check on a street grid

We checked these results on a grid of 264 streets in which every fourth street is an arterial ridden at twice the speed of the rest. Each draw takes 30 streets at random; the rider goes to the nearest unvisited sampled street by travel time, rides along it, and repeats. Running the draw and the routing 4,000 times gives m for every street. Arterials have 2.6 times the traversal intensity of other streets on average, and the most-ridden street has 5 times the intensity of the least.

Across 2,000 further rides, the number of frames on each street matches the rate in Proposition 1: the total is 1.002 times the predicted total, and the correlation across streets is 0.97. We then set the outcome to be hard for road-class reweighting: within each class, the streets the rule favors are busier and more male.

Table 1: Estimates of women's share of pedestrians on a simulated street grid where the true share is 22.2% and, within each road class, streets the routing rule favors are more male. Each row summarizes 2,000 rides. Design weights divide each frame's speed by the street's traversal intensity, computed from 4,000 separate runs of the draw and the routing rule. Bias is the mean minus the truth; SD is the standard deviation of the estimates.

	Mean (%)	Bias (pp)	SD (pp)
Per minute	20.9	−1.3	2.5
Per metre (speed weights)	19.7	−2.5	2.5
Speed weights, reweighted by road class	21.3	−0.9	2.5
Design weights (speed over traversal intensity)	22.3	+0.1	2.6

Table 1 matches Proposition 2. Speed weights alone are 2.5 points low. Reweighting by road class removes most of that but stays 0.9 points low, because it cannot see which streets within a class the rule favors. Design weights are within 0.1 points of the truth.

5 Estimation

5.1 Weighted totals, ratios, and means

Let w_i be the weight of sampled unit i. For a point sample, $w_i = 1/\pi_i$. The Horvitz–Thompson estimator of a total,

$$\widehat{T}_Y = \sum_{i \in \mathcal{S}} w_i Y_i, \tag{7}$$

is unbiased for $T_Y = \sum_{i \in \mathcal{U}} Y_i$ over repeated draws [Horvitz and Thompson, 1952]. The ratio and the mean are estimated by

$$\widehat{R} = \frac{\sum_{i \in \mathcal{S}} w_i Y_i}{\sum_{i \in \mathcal{S}} w_i Z_i}, \qquad \bar{Y} = \frac{\sum_{i \in \mathcal{S}} w_i Y_i}{\sum_{i \in \mathcal{S}} w_i}. \tag{8}$$

The threshold share is the mean of $\mathbb{1}\{Y_i \geq \tau\}$. With equal-probability sampling the weights cancel and these are the sample ratio and the sample mean.

Proposition 3 (Ratio estimator)**.** *If every unit has a known positive inclusion probability and measurement returns fixed values (Y_i, Z_i), then $\widehat{R}$ is consistent for R with bias of order n^{-1}, and its variance is estimated to first order by the variance of the weighted total of $e_i = (Y_i - \widehat{R} Z_i)/\widehat{T}_Z$.*

The bias arises because $\widehat{R}$ divides one random total by another [Cochran, 1977].

5.2 Variance

Write $c_i = w_i e_i$ for a ratio and $c_i = w_i(Y_i - \bar{Y})/\sum_j w_j$ for a mean. Sum c_i within each independent unit g (a segment in a point sample, a day or an itinerary in a route sample) to get C_g. With G_h independent units in stratum h,

$$\widehat{\mathbb{V}} = \sum_h \frac{G_h}{G_h - 1} \sum_{g \in h} \left(C_g - \bar{C}_h \right)^2 . \tag{9}$$

With one stratum and each row its own unit, this is the usual variance of a ratio or a mean under simple random sampling; with days or itineraries as units, it is the cluster-robust variance [Cameron and Miller, 2015]. It treats units as drawn with replacement and so omits the finite population correction. That overstates the variance by the sampling fraction. The fraction is about 1% or less in the Street View samples of Indian cities. It may be as large as a sixth in the major-road samples (Bangkok's 1,000 sampled segments carry identifiers up to 6,267), so those standard errors are somewhat conservative. We omit the correction throughout because the size of the frame was not saved with most of the samples; the sampler now saves it.

After post-stratification, c_i is replaced by its departure from the weighted mean of c in its road class. Fixing the class shares removes the variation between classes, and only variation within classes remains. This treats the shares as known. Ours are estimated from the drawn sample, which adds variance the formula leaves out; the simulation below estimates shares the same way and finds the intervals still cover at the stated rate.

Intervals for clustered samples use a t distribution with degrees of freedom equal to the number of clusters minus the number of strata; intervals for point samples, with hundreds of independent segments, use the normal distribution. The video samples have 9 to 13 days each, and intervals based on so few clusters tend to be too narrow [Cameron and Miller, 2015]. Treating days as independent is itself an approximation: consecutive days share a season and a rider.

All of this is implemented in the `geoestimate` package. Its weighted, stratified, clustered, and post-stratified estimates and standard errors agree with those of R's `survey` package to eight digits on test data.

5.3 Coverage

A sample is complete only when every drawn unit is either observed or recorded as missing. If units are missing at different rates in different road classes, post-stratifying the observed units to the drawn sample's class shares corrects for it, on the assumption that within a class

the missing units resemble the observed ones. If that assumption fails, the estimate describes covered streets.

6 A Check Against a Known Truth

We built an artificial city of 20,000 segments in three road classes. Classes differ in women's share of pedestrians, in crowding, and in speed, and within a class more crowded segments are slower. The true share of women among pedestrians is 24.7%. We drew two kinds of sample a thousand times each. The point sample observes 600 segments drawn at random. The route sample rides 150 segments on each of 12 days, half of them on major roads though major roads are 8% of the city, and saves a frame every 60 seconds; each day also shifts the share of women a little, so frames from one day are correlated. As in the applications, the road-class shares used for reweighting are estimated from a separate draw of 1,000 segments.

Table 2: Estimates of women's share of pedestrians in a simulated city where the true share is 24.7%. Each row summarizes 1,000 samples. "Per minute" weights frames equally; "per metre" weights each frame by speed; "road classes only" post-stratifies road classes to their shares in a drawn sample of 1,000 segments without speed weights; "road network" does both. Mean and SD are the mean and standard deviation of the estimates across samples, bias is the mean minus the truth, SE is the mean estimated standard error, and coverage is the share of nominal 95% intervals that contain the truth.

	Mean (%)	Bias (pp)	SD (pp)	SE (pp)	Coverage (%)
Point sample	24.8	0.0	1.4	1.4	95
Route sample, per minute	22.1	−2.6	1.0	1.0	32
Route sample, per metre	19.8	−4.9	1.0	0.9	1
Route sample, road classes only	24.8	+0.1	1.1	1.1	95
Route sample, road network	24.9	+0.1	1.2	1.1	95

Table 2 gives the results. The point sample is unbiased and its intervals cover the truth 95% of the time. The route sample taken at face value is 2.6 points too low and its intervals cover 32% of the time. Weighting by speed alone is worse, 4.9 points too low, because it moves weight from slow residential streets to fast major roads, which the route already over-represents. Post-stratifying to road classes removes the bias, with or without speed weights, and the intervals then cover 95% of the time.

Two things about this result limit what it shows. Post-stratification alone does as well as the combination because, in this city, women's share does not vary with speed within a road class. Speed weights matter for quantities that do, such as the number of pedestrians per frame. And the simulation gives post-stratification its best case: road class is the only way

the route differs from the city. Real routes may also favor particular streets within a class, and the simulation says nothing about how the estimator does then.

7 Applications

7.1 Road condition from Street View

This is the point sample in its plainest form. In late 2017 we drew 1,000 segments each of primary, secondary, tertiary, and trunk road in Bangkok, Dhaka, Jakarta, and Lagos, and 5,000 in Wayne County, Michigan, and requested Street View images at the start of each. Workers on Amazon Mechanical Turk coded each image for potholes, cracks, litter, a paved sidewalk, and clear road markings.

Table 3: Condition of major roads from Street View images at sampled road segments. Entries are percentages of segments. The frame is primary, secondary, tertiary, and trunk roads cut into 500-meter segments; "Covered" is the share of sampled segments for which Street View returned an image. Estimates are post-stratified to the drawn sample's mix of road classes.

	Sampled	Covered (%)	Potholes	Cracks	Litter	Paved sidewalk	Clear markings
Bangkok	1,000	100	6	24	6	51	83
Jakarta	1,000	98	23	44	21	49	35
Lagos	1,000	87	6	20	15	30	24
Wayne County	5,000	97	7	62	9	67	66

Coverage was 24.6% in Dhaka, which we therefore dropped, 87% in Lagos, and 97% or more elsewhere (Table 3). 23% of segments in Jakarta show potholes, against 6% to 7% in the other three places; standard errors across the table run from 0.4 to 1.6 percentage points. Adjusting for the road classes of the uncovered segments changes no estimate by more than 0.6 points. The estimates are about the streets as they were when photographed, which ranged over several years, and about what one image at a segment's start shows. They rest on workers' judgments, which we did not validate against a second coding here.

7.2 Women's presence from Street View

The second point sample uses a frame that adds residential and unclassified roads. Of 2,500, 2,000, and 2,500 locations drawn in Mumbai, Navi Mumbai, and Delhi, 1,500 in each city were checked for Street View coverage; 83%, 76%, and 78% had a panorama. Coders counted women and men on foot in images from 498, 437, and 500 locations. Neither how the 1,500 were chosen from those drawn nor how the coded locations were chosen from the covered ones

was recorded. About a third of the checked sample was observed, and the estimates rest on the assumption that neither choice, nor coverage itself, was related to who was on the street.

Women are 20.8% of pedestrians in Mumbai, 25.0% in Navi Mumbai, and 20.2% in Delhi (Table 4, last column). Adjusting for road class moves each by less than half a point.

Platform imagery has properties that matter for an outcome like this one. Google blurs faces and license plates [Google, 2026a, Frome et al., 2009], which protects the people photographed and makes some of them harder to classify. The platform, not the researcher, chooses when to photograph, and its vehicles work in daylight. And images can be years old, with dates that vary within a city [Curtis et al., 2013]. Street View is better suited to things that change slowly, such as sidewalks, than to who is on the street at a given hour.

7.3 Women's presence from video

A rider with a chest-mounted camera rode itineraries built from 1,000 sampled points in each of Mumbai, Bangalore, and Delhi and 10,000 in Navi Mumbai, over 11, 9, 13, and 14 days. Coders counted women and men on foot in 9,240 frames. Counts above ten were recorded as eleven, which affects men's counts in 9.4% of Mumbai frames and under 4% elsewhere, and so overstates women's share in crowded Mumbai frames by an amount we cannot recover. This is a route sample, and Table 4 and Figure 2 report it three ways. All three use the frames for which the camera's track gives a speed: 2,787, 1,986, 1,968, and 2,196 frames, the last on 13 of Delhi's 14 days.

Table 4: Women's share of pedestrians (%), with 95% confidence intervals. The first three columns use video frames. "Per minute of riding" weights frames equally; "per metre of route" weights each frame by the rider's speed; "road network" also post-stratifies road classes to the drawn sample. All three use the frames with a GPS speed, and their intervals treat fieldwork days as clusters. "Street View" uses images at sampled locations, post-stratified to the drawn sample's road classes, with locations as independent units. Street View was not collected in Bangalore.

	Per minute of riding	Per metre of route	Road network	Street View
Mumbai	19.1 [17.4, 20.7]	19.0 [17.4, 20.6]	20.1 [17.9, 22.2]	20.8 [18.8, 22.9]
Navi Mumbai	18.0 [15.6, 20.3]	17.8 [15.8, 19.7]	20.5 [15.7, 25.3]	25.0 [21.3, 28.7]
Delhi	20.4 [17.7, 23.1]	18.7 [14.2, 23.1]	23.5 [18.9, 28.2]	20.2 [17.3, 23.0]
Bangalore	27.5 [25.9, 29.1]	25.7 [23.1, 28.2]	27.6 [24.4, 30.9]	–

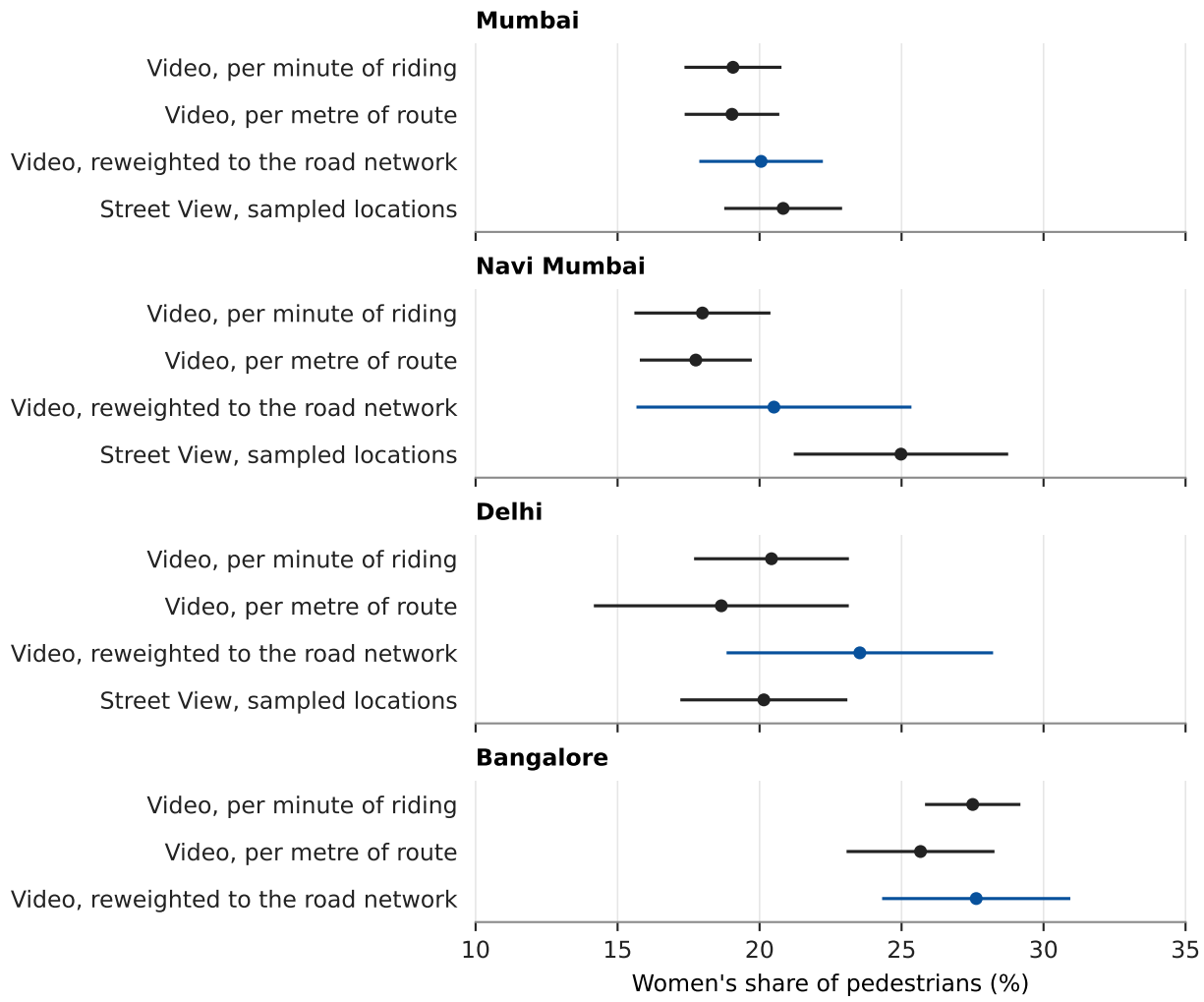


Figure 2: Women's share of pedestrians under each weighting of the video frames, and from Street View. Points are estimates and lines 95% confidence intervals, computed as in Table 4. The estimate reweighted to the road network is in blue.

Counted per minute of riding, women are 19.1% of pedestrians in Mumbai, 18.0% in Navi Mumbai, 20.4% in Delhi, and 27.5% in Bangalore. Weighting by speed lowers the Delhi and Bangalore estimates by about two points. Reweighting to the road network raises them again: to 23.5% in Delhi and 27.6% in Bangalore.

The reason is visible within road classes. In Delhi women are 24.9% of pedestrians in the 582 frames on residential streets and 17.6% in the 981 frames on secondary and larger roads, and the route over-represents the larger roads (Figure 1). As in the simulation, road class does most of the work: post-stratifying without speed weights gives 24.1% in Delhi. Speed weights matter more for quantities that vary with speed within a class. Delhi frames show 1.24 pedestrians each counted per minute and 0.90 per metre.

How much the reweighting moves the estimate is itself uncertain. Because both estimates use the same frames, we resample days and recompute both. In Delhi the reweighted estimate is 3.1 points above the per-minute one, with a 95% interval of 0.5 to 6.8. The shifts are 2.5 points in Navi Mumbai (0.5 to 8.1), 0.9 in Mumbai (0.1 to 1.8), and 0.1 in Bangalore (-2.7 to 3.0). With 9 to 13 days to resample, these intervals are rough. They say the reweighting raises the estimate in Delhi and Navi Mumbai, by an amount that could be half a point or several. Changing the floor on speed from 0.5 to 0.1 or 1.0 meters per second moves no reweighted estimate by more than 0.1 points.

The adjustments also change how much the data can say. In Delhi the interval is already about 70% wider once frames are weighted by speed, because a few fast frames carry much of the weight, and it stays that wide after reweighting. The clustered standard errors are between 0.6 and 1.9 times the ones that treat frames as independent. In Bangalore they are smaller than the independent ones, which is a sign that 13 days are too few to estimate the between-day variance and that Bangalore's intervals are too narrow.

Across cities and weightings the estimates run from 17.8% to 27.6%. The Street View estimate is within two points of every video estimate in Mumbai and inside the range of the video estimates in Delhi. They are furthest apart in Navi Mumbai, where Street View gives 25.0% and video 20.5% at most. The two sources differ in year, hour, and the roads they cover, and we cannot say which difference is responsible. The location mean L is within 1.1 points of R in every city, so crowded places are not notably more male than quiet ones.

7.4 Do routes favor particular streets within a road class?

Reweighting by road class is right when, within a class, the streets the route uses resemble the rest (Proposition 2). The video allows a direct check. Some frames fall within 100 meters of a sampled point, on or beside a street the draw chose. The rest are on the way between points, on streets the routing chose. If routing favors a different kind of street within a class, the two groups should differ.

Table 5: Women's share of pedestrians in video frames within 100 meters of a sampled point (near) and farther away (far), by road class. The difference is near minus far, with a 95% interval from resampling fieldwork days. Navi Mumbai is omitted: its 10,000 sampled points leave almost no frame far from one.

	Road class	Frames near	Frames far	Near (%)	Far (%)	Difference (pp)
Mumbai	Secondary and above	834	630	18.6	18.4	+0.2 [−2.4, +2.2]
Mumbai	Residential	431	393	19.4	21.6	−2.2 [−6.3, +1.8]
Mumbai	Tertiary	240	221	19.3	17.7	+1.6 [−3.2, +5.1]
Delhi	Secondary and above	51	968	13.9	17.1	−3.2 [−13.7, +7.0]
Delhi	Residential	188	413	25.6	23.6	+2.0 [−6.4, +8.0]
Delhi	Tertiary	62	525	18.9	19.7	−0.8 [−9.8, +10.2]
Bangalore	Secondary and above	50	808	29.6	26.3	+3.3 [−13.8, +21.1]
Bangalore	Residential	142	340	26.5	31.3	−4.8 [−10.8, +2.9]
Bangalore	Tertiary	41	506	14.1	26.2	−12.2 [−20.6, +1.4]

Table 5 shows no such difference. Across 9 city-by-class cells the difference ranges from −12.2 to +3.3 points, the frame-weighted mean is −1.4 points, and 0 of the intervals exclude zero. The check is not sharp. The intervals are 2 to 17 points wide on either side, so it can rule out large within-class selection but not a few points of it. A second comparison mostly points the same way. Class by class, Street View at sampled locations and video on routes give shares within three points of each other in Mumbai and Delhi, except on Delhi's tertiary roads, where video is eight points higher.

7.5 Noise in Delhi

Between 28 April and 16 May 2026 a rider carried a sound meter along 56 of the 85 Delhi itineraries, on 15 days. Many of these are the itineraries the video covers. At stops along each itinerary the rider recorded the lowest and highest level the meter showed during the reading: 381 readings at planned stops and 81 more while halted in traffic. We do not know how the 56 itineraries ridden were chosen from the 85; they are spread across the city much as the rest are.

The readings are a route sample. Half of the planned-stop readings (50%) are on residential roads, against 85% of sampled points. Table 6 reports the readings as taken and reweighted to the sample's road classes, with days as clusters. Where on an itinerary to stop was the rider's choice. If the rider stopped where stopping was easy, the readings differ from the streets of their class in ways the reweighting does not touch.

Table 6: Street noise in Delhi from sound-meter readings, with 95% confidence intervals. Each reading gives the minimum and maximum level during the reading. The first two columns use readings at planned stops: "as read" weights them equally, and "road network" post-stratifies them to the road classes of the drawn sample. Traffic stops are readings taken while halted in traffic and are not part of the other columns. Intervals treat fieldwork days as clusters. Every traffic-stop reading had a minimum above 55 dB, so that cell has no interval.

	As read	Road network	Traffic stops
Mean minimum (dB)	64.5 [63.1, 66.0]	62.4 [60.8, 64.0]	71.5 [70.2, 72.8]
Mean maximum (dB)	82.3 [78.8, 85.8]	78.3 [75.1, 81.5]	94.8 [91.9, 97.6]
Minimum at or above 55 dB (%)	87.4 [81.6, 93.2]	83.5 [75.4, 91.6]	100.0
Maximum at or above 100 dB (%)	14.4 [7.7, 21.2]	8.1 [3.1, 13.0]	33.3 [25.8, 40.9]

Reweighted to the road network, the mean minimum is 62.4 dB and the mean maximum 78.3 dB. In 83.5% of readings even the quietest moment is at or above 55 dB, the daytime standard for residential areas. The reweighting matters most for the loudest moments: 14.4% of readings as taken peak at or above 100 dB, but 8.1% once residential streets get their due weight. The two intervals overlap, so the size of this drop is uncertain; its direction follows from residential streets being quieter. Readings in traffic are louder, with a mean maximum of 94.8 dB and a third peaking at or above 100 dB. They describe a condition, being stopped in traffic, and not a sample of streets, so we keep them apart.

A spot reading of a few seconds is not the day-long average that the standard is written for, and we did not calibrate the meter against a reference instrument. The comparison with 55 dB locates the readings; it is not a finding of legal exceedance.

7.6 Air quality in Delhi

On 5 of the days the rider also read a hand-held air monitor, 98 times in all. Mean PM2.5 was 118 μg/m^3 (95% interval 80 to 156, days as clusters). We report it as read. With 98 readings on 5 days, the adjustments above would rest on a handful of readings per road class, and the interval is already wide. This application shows the limit of a small route sample more than it establishes a level.

8 Ethics

The imagery here shows people in public who did not agree to be recorded, and several features of the work are meant to limit what that costs them.

What is recorded about people. Coders count people and record apparent gender from

clothing, hair, and build. No face recognition is used, no individual is identified or tracked across frames, and the analysis data contain counts only. Apparent gender is a coder's judgment about appearance. It imposes two categories on people who may not fit them.

What is kept and released. Video and full-resolution frames can identify people and are not released. Coders saw reduced-resolution frames. The public data are counts with a location and a time. Street View images are blurred by Google before we receive them, and we release no images from either source.

Location and the field team. The rider's routes are in the data as the positions of frames and readings. We checked that each day's first and last positions are scattered across the city and do not mark a home. The rider's route plan, which does contain a home address, is not released. Riding a motorbike in city traffic for weeks carries real risk to the rider, which a study of this kind should weigh against what it learns.

What the method could be used for. A procedure that samples streets and counts who is on them could be turned to watching particular places or groups. Probability sampling pulls the other way: it spreads observation thinly across a city and makes no place the target. We would not extend the method to identify individuals, to follow people, or to map the presence of a group in order to act against it.

Platform terms. Street View images are used under Google's terms, which restrict storing and redistributing images; we store counts and metadata.

Review. The video study was reviewed by the University of Southern California Institutional Review Board and deemed exempt (UP-25-00545).

9 Beyond Streets

Nothing above depends on the unit being a street. The frame can be any list of places that can be enumerated and reached: grid cells, parcels, transit stops, polling stations, water points. The two designs carry over. If the instrument observes the drawn place, the draw's probabilities apply. If the instrument observes the way to the place, the observations sample the way, and what must be asked is what governs how often each kind of place is seen (time, distance, access) and whether a recorded quantity, as speed was here, can undo it.

Roads are a convenient case because one object serves three purposes: the road network is the frame, the means of travel, and, for many outcomes, the thing measured. That convenience is also why route samples of roads go wrong in a patterned way. Travel concentrates on the roads built for travel.

10 Limitations

Road class is a coarse account of how routes differ from the network. Within residential streets, a route uses those that connect, and connecting streets may be busier than dead ends. The grid simulation shows the bias this leaves when it happens. In the video, frames near sampled points and frames far from them do not differ detectably within road classes (Table 5), but that check cannot exclude a difference of a few points.

The road-class shares used for reweighting come from the drawn sample and have their own sampling error, which the intervals ignore. In Delhi only 4% of sampled points are on secondary or larger roads, so that share is itself imprecise.

Fieldwork covered daytime on some days of some months. Every estimate is about those hours. Who is on the street at night, and how loud the street is then, are not measured.

The number of independent units is small for the video (9 to 13 days per city) and the sensors (15 days for noise, 5 for air), and intervals with so few clusters are too narrow by an amount we have not estimated.

Several things were not recorded when the data were collected and limit what can be claimed: the size of the frame, how checked and coded Street View locations were chosen, which video frames were annotated in Mumbai and Navi Mumbai, and how the itineraries actually ridden were chosen. The sampler now saves each segment's inclusion probability, stratum, and length with the sample.

11 Conclusion

A random draw from the road network gives street-level measurement a population to describe. Whether the measurements describe it depends on a second thing, which is easy to overlook: whether the instrument observed the units that were drawn. When it did, as with images requested at sampled segments, the usual survey estimators apply and the work is in reporting coverage. When the instrument rode between the units, the data describe the ride. In our Delhi fieldwork 45% of frames are on large roads that make up 4% of the sample.

A route sample is still a probability sample. The draw and the routing rule fix how often the rider passes each street, the timer and the speed fix how often a frame falls there, and weighting a frame by speed over traversal intensity is unbiased without a model. Speed weights alone are right only when the streets the rule favors are like the rest, and on a road network they are not: the rule favors large roads. Reweighting by road class approximates the full weights by treating traversal intensity as constant within a class. In our video the approximation looks adequate, since frames near sampled points and frames on the way do

not differ detectably within classes. It moves the share of women on Delhi's streets up by 3.1 points, with an interval from 0.5 to 6.8, and lowers the share of very loud noise readings from 14.4% to 8.1%.

The better course is to compute the traversal intensity itself, which requires that the route be a rule that can be rerun. Ours was a navigation application, and we could not.

For new studies the simpler course is to keep the two designs apart. Observe at the sampled points when the question is about the network. Record continuously when the question is about travel, or when the routing rule is saved so that its traversal intensity can be computed. The same sample can also support comparisons over time or between places if the protocol is repeated, though we have not done that here.

References

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.

William G. Cochran. *Sampling Techniques*. John Wiley & Sons, New York, 3 edition, 1977. ISBN 978-0-471-16240-7.

Jacqueline W. Curtis, Andrew Curtis, Jennifer Mapes, Andrea B. Szell, and Adam Cinderich. Using Google Street View for systematic observation of the built environment: Analysis of spatio-temporal instability of imagery dates. *International Journal of Health Geographics*, 12(1):53, 2013. doi: 10.1186/1476-072X-12-53.

Lorenzo Fattorini. Applying the Horvitz-Thompson criterion in complex designs: A computer-intensive perspective for estimating inclusion probabilities. *Biometrika*, 93(2):269–278, 2006. doi: 10.1093/biomet/93.2.269.

Andrea Frome, German Cheung, Ahmad Abdulkader, Marco Zennaro, Bo Wu, Alessandro Bissacco, Hartwig Adam, Hartmut Neven, and Luc Vincent. Large-scale privacy protection in Google Street View. In *2009 IEEE 12th International Conference on Computer Vision*, pages 2373–2380. IEEE, 2009. doi: 10.1109/ICCV.2009.5459413.

Dustin Fry, Stephen J. Mooney, Daniel A. Rodríguez, Waleska T. Caiaffa, and Gina S. Lovasi. Assessing Google Street View image availability in Latin American cities. *Journal of Urban Health*, 97(4):552–560, 2020. doi: 10.1007/s11524-019-00408-7.

Timnit Gebru, Jonathan Krause, Yilun Wang, Duyun Chen, Jia Deng, Erez Lieberman Aiden, and Li Fei-Fei. Using deep learning and Google Street View to estimate the demographic makeup of neighborhoods across the United States. *Proceedings of the National Academy of Sciences*, 114(50):13108–13113, 2017. doi: 10.1073/pnas.1700035114.

Google. Google-contributed Street View imagery policy, 2026a. URL https://www.google.com/streetview/policy/. Undated web page. Accessed: 2026-11-12.

Google. Street view image metadata, 2026b. URL https://developers.google.com/maps/documentation/streetview/metadata. Street View Static API documentation, Google Maps Platform. Last updated 2026-11-09. Accessed: 2026-11-12.

Google. Street view static API overview, 2026c. URL https://developers.google.com/maps/documentation/streetview/overview. Google Maps Platform documentation. Last updated 2026-11-09. Accessed: 2026-11-12.

D. Holt and T. M. F. Smith. Post stratification. *Journal of the Royal Statistical Society. Series A (General)*, 142(1):33–46, 1979. doi: 10.2307/2344652.

D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952. doi: 10.1080/01621459.1952.10483446.

Cheryl M. Kelly, Jeffrey S. Wilson, Elizabeth A. Baker, Douglas K. Miller, and Mario Schootman. Using Google Street View to audit the built environment: Inter-rater reliability results. *Annals of Behavioral Medicine*, 45(Suppl 1):S108–S112, 2013. doi: 10.1007/s12160-012-9419-9.

Sharon L. Lohr. *Sampling: Design and Analysis*. Chapman and Hall/CRC, 3 edition, 2021. doi: 10.1201/9780429298899.

Ministry of Environment and Forests, Government of India. The noise pollution (regulation and control) rules, 2000, 2000. URL https://cpcb.gov.in/displaypdf.php?id=Tm9pc2UtU3RhbmRhcmRzL25vaXNlX3J1bGVzXzIwMDAucGRm. Notification S.O. 123(E), New Delhi, 14 February 2000, under the Environment (Protection) Act, 1986. Consolidated text as amended, hosted by the Central Pollution Control Board. Accessed: 2026-11-12.

Nikhil Naik, Jade Philipoom, Ramesh Raskar, and Cesar Hidalgo. Streetscore – predicting the perceived safety of one million streetscapes. In *2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 793–799. IEEE, 2014. doi: 10.1109/CVPRW.2014.121.

OpenStreetMap contributors. OpenStreetMap, 2026. URL https://www.openstreetmap.org. Data available under the Open Database License. Accessed: 2026-11-12.

Shilpa Phadke, Sameera Khan, and Shilpa Ranade. *Why Loiter? Women and Risk on Mumbai Streets*. Penguin Books, New Delhi, 2011. ISBN 978-0-14-341595-4.

Andrew G. Rundle, Michael D. M. Bader, Catherine A. Richards, Kathryn M. Neckerman, and Julien O. Teitler. Using Google Street View to audit neighborhood environments. *American Journal of Preventive Medicine*, 40(1):94–100, 2011. doi: 10.1016/j.amepre.2010.09.034.

Robert J. Sampson and Stephen W. Raudenbush. Systematic social observation of public spaces: A new look at disorder in urban neighborhoods. *American Journal of Sociology*, 105(3):603–651, 1999. doi: 10.1086/210356.

Amartya Sen. More than 100 million women are missing. *The New York Review of Books*, 37(20):61–66, December 1990. URL https://www.nybooks.com/articles/1990/12/20/more-than-100-million-women-are-missing/.

J. G. Wardrop. Some theoretical aspects of road traffic research. *Proceedings of the Institution of Civil Engineers*, 1(3):325–362, 1952. doi: 10.1680/ipeds.1952.11259.

World Health Organization. Compendium of WHO and other UN guidance on health and environment, 2022 update. Technical Report WHO/HEP/ECH/EHD/22.01, World Health Organization, 2022. URL https://iris.who.int/handle/10665/352844.

World Health Organization. Regional Office for Europe. *Environmental Noise Guidelines for the European Region*. World Health Organization. Regional Office for Europe, 2018. ISBN 978-92-890-5356-3. URL https://iris.who.int/handle/10665/279952.